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E23CSEU2189

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# CSET419 — Lab 7
# Neural Style Transfer using VGG19
# Content: CIFAR-10
# Style: WikiArt
# ======================================

import torch
import torch.optim as optim
from torchvision import models, transforms, datasets
from PIL import Image
import matplotlib.pyplot as plt

device = torch.device("cuda" if torch.cuda.is_available() else "cpu"

# --------------------------------------
# Image size
# --------------------------------------

image_size = 256

transform = transforms.Compose([
    transforms.Resize((image_size, image_size)),
    transforms.ToTensor()
])

# --------------------------------------
# Load CIFAR-10 Content Image
# --------------------------------------

cifar = datasets.CIFAR10(
    root="./data",
    train=True,
    download=True
)

content_img, label = cifar[100]   # choose index

content_img = content_img.resize((256,256), Image.BICUBIC)

content_img = transforms.ToTensor()(content_img).unsqueeze(0).to(dev
```

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content_img = transforms.ToTensor()(content_img).unsqueeze(0).to(dev

# ---------------------------------------
# Load WikiArt Style Image
# ---------------------------------------

style_path = "wikiart_style.jpg"   # place style image in folder

style_img = Image.open(style_path).convert("RGB")

style_img = transform(style_img).unsqueeze(0).to(device)

# ---------------------------------------
# Display Images
# ---------------------------------------

def show(img, title):

    img = img.clone().cpu().squeeze(0)
    img = transforms.ToPILImage()(img)

    plt.imshow(img)
    plt.title(title)
    plt.axis("off")
    plt.show()

show(content_img,"Content Image (CIFAR10)")
show(style_img,"Style Image (WikiArt)")

# ---------------------------------------
# Load Pretrained VGG19
# ---------------------------------------

vgg = models.vgg19(pretrained=True).features.to(device).eval()

for param in vgg.parameters():
    param.requires_grad = False

# ---------------------------------------
# Feature Extraction
# ---------------------------------------

def get_features(image, model):

    layers = {
        '0':'conv1_1',
        '5':'conv2_1',
        '10':'conv3_1',
        '19':'conv4_1',
```

```
            '21':'conv4_2',
            '28':'conv5_1'
        }

        features = {}
        x = image

        for name, layer in model._modules.items():

            x = layer(x)

            if name in layers:
                features[layers[name]] = x

        return features

    # ---------------------------------------
    # Gram Matrix (Style Representation)
    # ---------------------------------------

    def gram_matrix(tensor):

        b, c, h, w = tensor.size()

        tensor = tensor.view(c, h*w)

        gram = torch.mm(tensor, tensor.t())

        return gram

    # ---------------------------------------
    # Extract Features
    # ---------------------------------------

    content_features = get_features(content_img, vgg)
    style_features = get_features(style_img, vgg)

    style_grams = {layer: gram_matrix(style_features[layer]) for layer i

    # ---------------------------------------
    # Generated Image
    # ---------------------------------------

    generated = content_img.clone().requires_grad_(True).to(device)

    # ---------------------------------------
    # Loss Weights
    # ---------------------------------------
```

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style_weight = 1e6
content_weight = 1

optimizer = optim.Adam([generated], lr=0.003)

# ---------------------------------------
# Training Loop
# ---------------------------------------

steps = 300

style_layers = ['conv1_1','conv2_1','conv3_1','conv4_1','conv5_1']

for step in range(steps):

    gen_features = get_features(generated, vgg)

    content_loss = torch.mean(
        (gen_features['conv4_2'] - content_features['conv4_2'])**2
    )

    style_loss = 0

    for layer in style_layers:

        gen_feature = gen_features[layer]

        gen_gram = gram_matrix(gen_feature)

        style_gram = style_grams[layer]

        _, c, h, w = gen_feature.shape

        style_loss += torch.mean((gen_gram - style_gram)**2)/(c*h*w)

    total_loss = content_weight*content_loss + style_weight*style_lo

    optimizer.zero_grad()
    total_loss.backward()
    optimizer.step()

    if step % 50 == 0:
        print("Step:", step, "Loss:", total_loss.item())

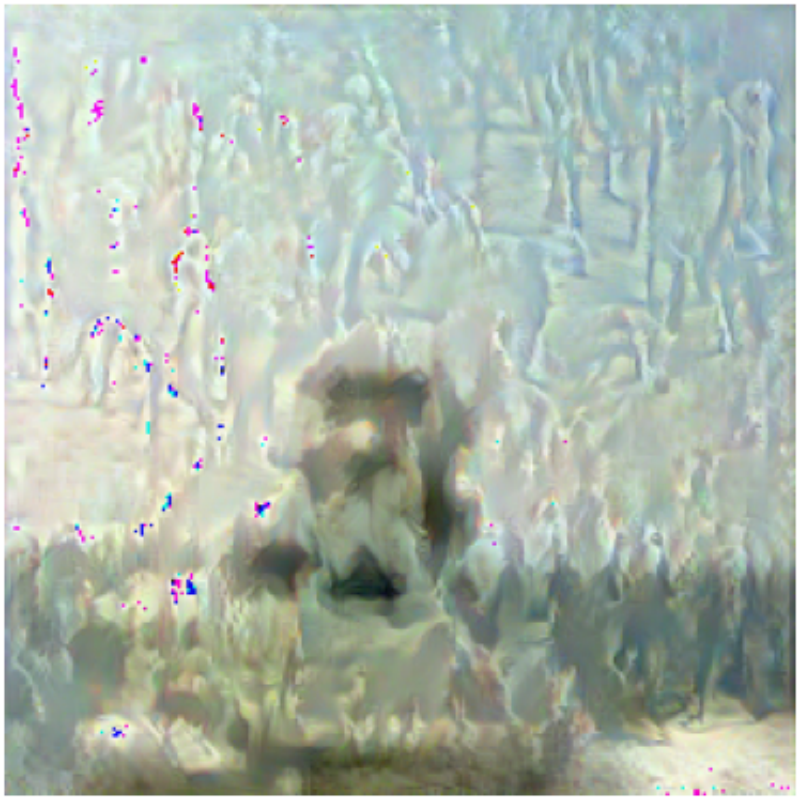
# ---------------------------------------
# Show Final Stylized Image
# ---------------------------------------
```

```
show(generated,"Stylized Image")
```

Content Image (CIFAR10)

Style Image (WikiArt)

```
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py
```

```
  warnings.warn(msg)
Downloading: "https://download.pytorch.org/models/vgg19-dcbb9e9d.pth
100%|██████████| 548M/548M [00:03<00:00, 173MB/s]
Step: 0 Loss: 5873502.0
Step: 50 Loss: 367362.0
Step: 100 Loss: 217917.609375
Step: 150 Loss: 169876.453125
Step: 200 Loss: 142909.0625
Step: 250 Loss: 124579.484375
```

Stylized Image